

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 15-03-2026**

Time: 02.00Pm to 05.00Pm **GTA-18** **Max. Marks: 180**

15-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2024-P2)_GTA-18_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. If $a+b+c=5, a^2+b^2+c^2=12$ and $a^3+b^3+c^3=25$ Then the value of $a^4+b^4+c^4$ is

A) $\frac{251}{6}$ **B)** $\frac{253}{6}$ **C)** $\frac{255}{6}$ **D)** $\frac{257}{6}$

2. If the line $x+y+1=0$ and $y-2x+5=0$ are tangents to a parabola whose focus at $(1,2)$, then the equation of normal to the parabola through $\left(\frac{29}{17},-\frac{14}{17}\right)$ is,

A) $2x+4y-1=0$ **B)** $y-3x+2=0$
C) $4x+y-5=0$ **D)** None of these

3. If $A(z_1), B(z_2), C(z_3)$ are vertices of a triangle such that $z_3=\left(\frac{z_2-iz_1}{1-i}\right) and |z_1|=3, |z_2|=4$ and $|z_2+iz_1|=|z_1|+|z_2|$, then area of triangle ABC is (sq. units)

A) 5 **B)** 25/2 **C)** 25/4 **D)** 5/2

4. A triangle ABC with vertices $A=(2,5), B=(5,2) and C=(-1,-1)$ inscribed on the curve $xy-x-y=3, H\equiv(\alpha,\alpha)$ is orthocenter. Let D,E and F be the feet of perpendiculars from A, B and C on BC,CA and AB respectively and inradius of ΔDEF is r, then $25r^2$ is___

A) 2 **B)** 4 **C)** 8 **D)** 16

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. Let $|r| < 1$ and $S_n = \sum_{k=1}^{\infty} k^n r^k$, then which of the following options is/are corrct ?

A) $S_0 = \frac{1}{1-r}$ **B)** $rS_1 = (S_0)^2$

C) $S_{n+1} = r\frac{d}{dr}(S_n)$ **D)** $rS_2 = S_0^2 + 2S_0^3$

6. In ΔABC, if $\angle B = \sec^{-1}\left(\frac{5}{4}\right) + \operatorname{cosec}^{-1}\sqrt{5}, \angle C = \operatorname{cosec}^{-1}\left(\frac{25}{7}\right) + \cot^{-1}\left(\frac{9}{13}\right) and\ c = 3.$

(All symbols used have their usual meaning in a triangle.)
Which of the following is/are correct ?

A) tanA, tanB, tanC are in arithmetic progression

B) The distance between orthocenter and centroid of triangle with sides $a^2, b^{\frac{4}{3}}$ and c is equal to $\frac{5}{3}$.

C) Area of ΔABC is irrational

D) Radius of escribed circle drawn opposite to vertex A is rational

7. An unbiased die, with faces numbered 1, 2, 3, 4, 5, 6 is thrown 6 times and the list of the 6 numbers showing up is noted. If $P(A_i)$ denotes the probability that only i number of numbers appear in the list out of numbers 1, 2, 3, 4, 5, 6 then which of the following is/are correct

A) $P(A_3) > P(A_4)$ **B)** $P(A_4) = \frac{23400}{6^6}$

C) $\sum_{i=1}^{6} P(A_i) < 1$ **D)** $\max(P(A_i)) = P(A_4)$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. Let p(x) be a fourth degree polynomial, with derivative $p'(x)$, Such that p(1) = p(3) = p(5) = $p'(7)$ = 0. If x is a real number such that p(x) = 0 and x is not equal to 1, 3 or 5, then 11x is_____

9. Let $f:R\to R$ be a continuous function which satisfies that identity $f(2x)=3f(x)\forall x\in R.$ If $\int_0^1 f(x)dx=1,$ then $\int_1^2 f(x)dx$ is_____

10. Let $f:R\to R$ be a differentiable function such that f(0) = 0, f(1) = 1 and $|f'(x)|<2\forall x\in R,$ if a and b are real numbers such that the set of possible values of $\int_0^1 f(x)dx$ is the open interval (a,b) then $4(b-a)$ is_____

11. If $lx+13y+mz+n=0$ is a plane through intersection of planes $2x+3y-z+1=0$ and $x+y-2z+3=0$ and is perpendicular to plane $3x-y-2z=4,$ then $\frac{l-n}{m}$ equals

12. If a curve represented by differential equation $(\tan^{-1}y-x)dy=(1+y^2)dx,$ passes through (0, tan1) and $\left(\frac{\pi}{3}-1,b\right),$ then value of $[10b]$ is (where [ ] denotes greatest integer function.

13. From the point P(h, k) tangents, PA and PB are drawn to the circle $x^2+y^2-2x-4y+1=0$, Such that slope of chord AB is 2, then the value of $h+2k$ is:

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

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PARAGRAPH-I

Consider a plane $P: 2x + y - z - 5 = 0$, a line $L_1 : \frac{x-3}{2} = \frac{y+1}{-3} = \frac{z-2}{-1}$ and a point A(3, -4,1). L_2 is a line passing through A intersecting L_1 and parallel to the plane P.

14. Distance of plane containing L_1 *and* L_2 from origin equals

15. Line L_1 intersect plane P at Q and xy – plane at R, then the volume of tetrahedron OAQR is, (where 'O' is origin)

PARAGRAPH-II

If locus of point P(z) in complex plane is $|z + z_1| + |z + z_2| = 4$, where A represents z_1 as (1, 0) and B represents z_2 as (-1, 0) and $Q(\omega)$ is moving point inside the locus of P(z) such that all internal angle bisectors of triangle of ΔPAB concurrent at Q. Then, answer the following questions if $|\omega - \omega_1| + |\omega - \omega_2| = 2$.

16. $|\omega_1| + |\omega_2|$ is equal to

17. If infimum value of $|\omega - z_1| + |\omega - z_2|$ is equal to m, then [m] is (where [.] denotes greatest integer part). (infimum is the greatest lower bound example infimum of the set (2,3] is 2)

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PHYSICS **Max Marks: 60**

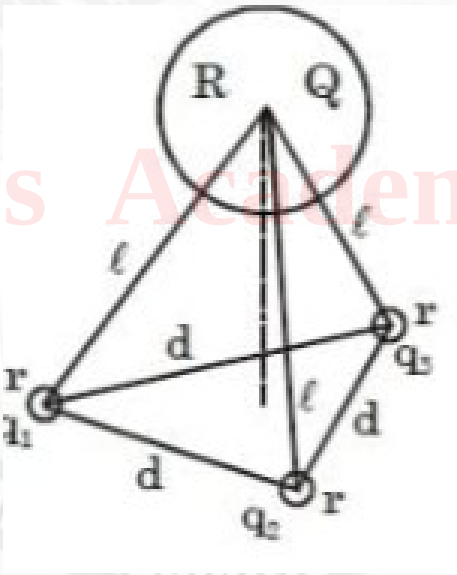
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

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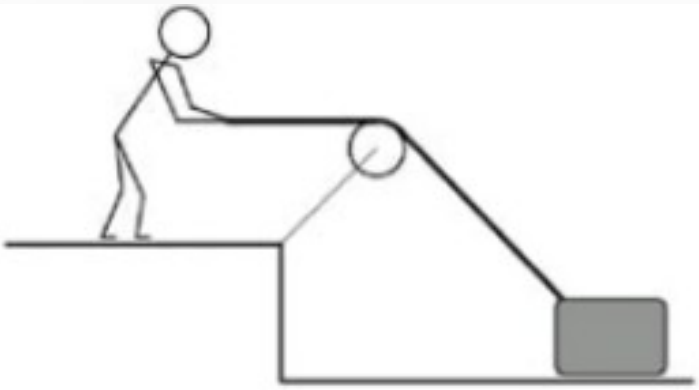
18. Three identical uncharged metal spheres are at the vertices of an equilateral triangle. One at a time, a small sphere is connected by a conducting wire with a large metal sphere that is charged. The centre of the large sphere is in the straight line perpendicular to the plane of equilateral triangle and passing through its centre (see figure). As a result, the first small sphere acquires charge q_1 and second charge. $q_2 (q_2 < q_1)$ The charge that the third sphere q_3, will acquire is : (Assume l >>R, l >>r, d >>R, d>>r)



A) $\frac{q_1^2}{q_2}$ **B)** $\frac{q_2^2}{q_1}$ **C)** $\sqrt{q_1 q_2}$ **D)** $\frac{q_1 + q_2}{2}$

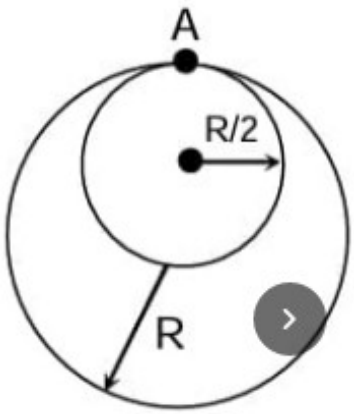
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19. As shown below, Lily is using the rope through a fixed light pulley to move a box with constant speed v. The kinetic friction coefficient between the box and the ground is $\mu < 1$; assume that the fixed pulley is massless and there is no friction between the rope and the fixed pulley. Then, while the box is moving, which of the following statement is correct ?



A) The magnitude of the force on the rope is constant.
B) The magnitude of friction between the ground and the box is decreasing
C) The magnitude of the normal force of the ground on the box is increasing
D) The pressure of the box on the ground is increasing.

20. Two rings made of same material and thickness, one of radius R and other of radius R/2 are welded together at point A. Now, it is hanged on a nail at wall, the nail touching both the rings at A. Now it is slightly displaced in the plane of rings and released. The period of small oscillations is



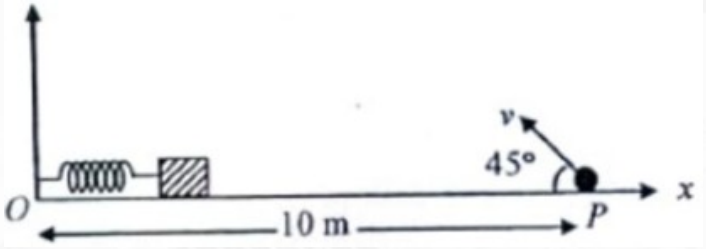
A) $2\pi\sqrt{\frac{2R}{5g}}$ **B)** $2\pi\sqrt{\frac{5R}{6g}}$ **C)** $2\pi\sqrt{\frac{9R}{5g}}$ **D)** $2\pi\sqrt{\frac{5R}{2g}}$

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21. A small block is connected to one end of a mass less spring of un-strecthed length 4.9 m. The other end of the spring (see the figure) is fixed. The system lies on a horizontal frictionless surface. The block is pulled by 0.2 m and released from rest at t = 0. It then executes simple harmonic motion with angular frequency $\omega = \frac{\pi}{3} rad/s.$

Simultaneously at t =0, a small pebble is projected with speed v from point P at an angle of 45° as shown in the figure. Point P is at a horizontal distance of 10 m from O. If the pebble hits the block at t = 1s, the value of v is

(take $g = 10 m/s^2$)



A) $\sqrt{50}\, m/s$ **B)** $\sqrt{51}\, m/s$ **C)** $\sqrt{52}\, m/s$ **D)** $\sqrt{53}\, m/s$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

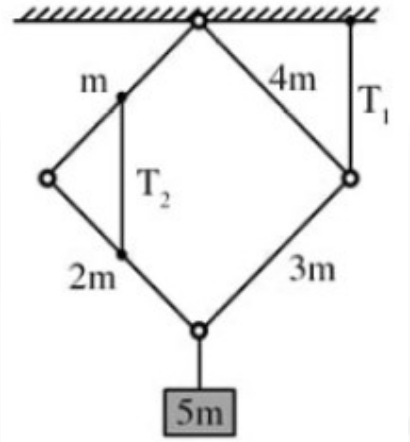
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
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For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. Four thin homogeneous rods of constant cross-section, having different masses m, 2m, 3m and 4m, but equal lengths, were joined by hinges to form the construction shown in the figure (the masses of the corresponding rods are indicated next to them). A load of mass 5m was suspended from the lower part of the construction. One thread connected the midpoints of the left rods with each other, and one thread connects one hinge with the ceiling as shown in the diagram. The right angle of the structure is fixed to the ceiling. The structure is in equilibrium, the strings are vertical. There is no friction in the system. The masses of the hinges and strings can be neglected

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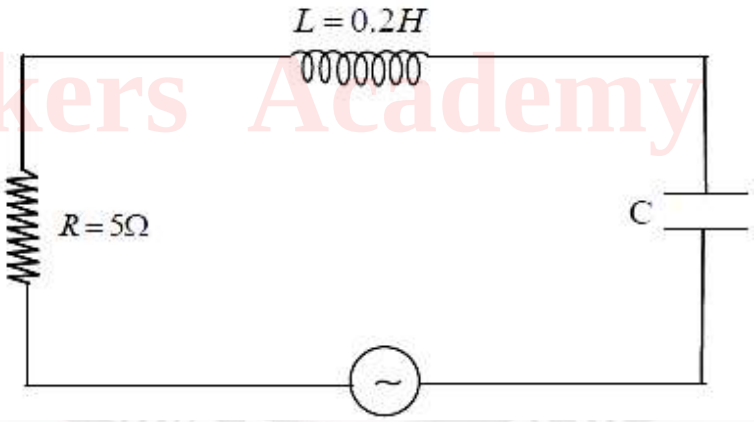
A) The value of T_1 is 2mg.

B) The value of T_2 is 13 mg

C) The value of hinge force due to ceiling is 13 mg

D) The value of hinge force due to ceiling is 17 mg.

23. An alternating voltage $E = 6\sin 20t + 8\cos 20t$ is applied to a series resonant circuit as shown. The correct statements are :



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A) The capacitance C is 12.5 mF

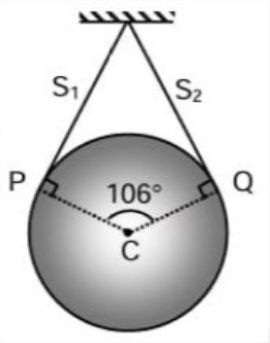
B) The resonant rms current in the circuit is $\sqrt{2}A$

C) Average power dissipated in the circuit is 10W

D) The quality factor of the current is 0.8

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24. A circular disc of mass 'm' and radius 'R' is held in equilibrium in a vertical plane by two identical strings 'S_1' and 'S_2' of negligible mass as shown in the figure. Choose the correct option(s).



A) Tension in the string S_2 when circular disc is in equilibrium is $\frac{5}{8}mg$.

B) Acceleration of point P just after cutting the string S_2 is $\frac{3}{5}g$.

C) Tension in the string 'S_1' immediately after cutting the string 'S_2' $is\frac{4mg}{15}$.

D) Tension in the string 'S_1' immediately after cutting the string 'S_2' is $\frac{8mg}{15}$

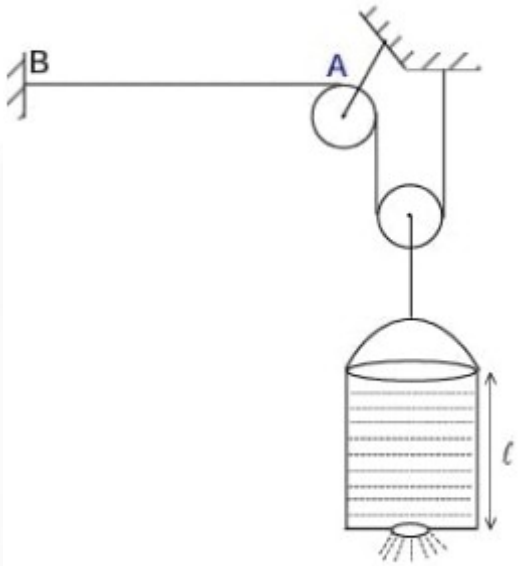
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
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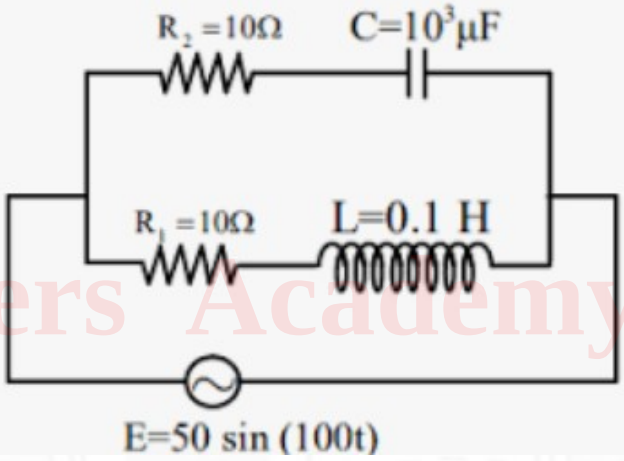
Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. In the given diagram, there is a cylindrical can of cross sectional area A and length $l = 0.4$ m completely filled with liquid of density 'ρ'. It is attached to massless pulley. A small hole of area 'a' = $\frac{A}{100}$ is made at the base of can. A wave pulse is generated at point A at an instant when small hole is made at base of can. When can gets empty, the wave pulse reaches point B. Then, find length AB (in meter) $\left(\frac{\rho A}{\mu}=\frac{1}{16}S.I\,unit\right)\mu$ is linear mass density of string 'AB' Neglect mass of can and thrust force due to efflux of liquid.

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26. For the circuit shown in the figure, find the peak current (in ampere) through the source

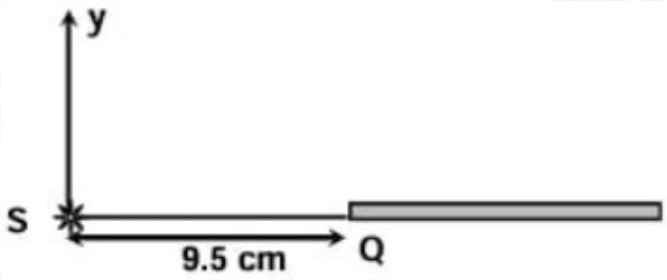


27. A container filled with air under pressure P_0 contains a soap bubble of radius R. The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S, P_0 is found to be $\frac{16nS}{R} SI\ unit..$

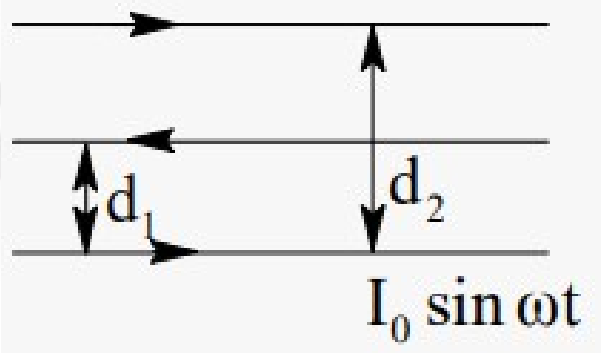
Find n.

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28. A string SQ is connected to a long heavier string at Q Linear mass density of the heavier string is 4 times of the string SQ. Length of SQ is 9.5 cm. Both the strings are subjected to same tension. A 50 Hz source connected at S produces transverse disturbance in the string. Wavelength of the wave in string SQ is observed to be 1 cm. If the source is switched on at time t = 0 and the smallest time (in sec) at which a point in the heavier string would oscillate in phase with the source at S is $\frac{T}{10}$. Find T.



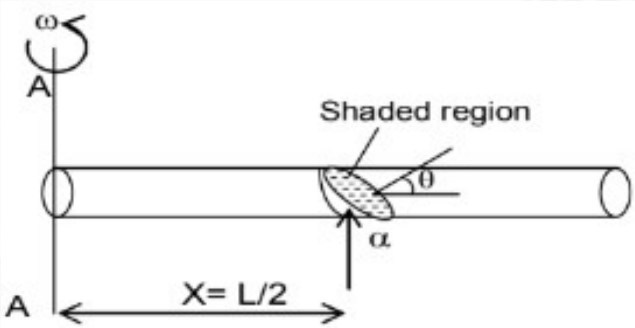
29. Two long straight wires of telephone circuit inside a house are coplanar with and run parallel to a third long straight conductor that carries a current $i = i_0 \sin \omega t$. The wires are at a distance of d_1 and d_2 from the conductor. The conductor induced a noise in the telephone circuit. The rms voltage per unit length of this induced emf is $x \times 5$, find x? Express your answer in μV. Take $d_2 = 2d_1, i_0 = \frac{2\sqrt{2}}{\ln 2} A, \omega = 50 rad/s$



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30. A thin uniform metallic rod of mass m, length L, Young modulus of elasticity Y and cross sectional area A is rotated by angular velocity ω about extreme and AA'. Consider a section on the rod at midpoint of rod. The normal stress in N/m^2 on the shaded region is $\frac{x}{4}$ then find x.

(Take the value of $\frac{M\omega^2 L\cos^2\theta}{A} = 10\,N/m^2$)



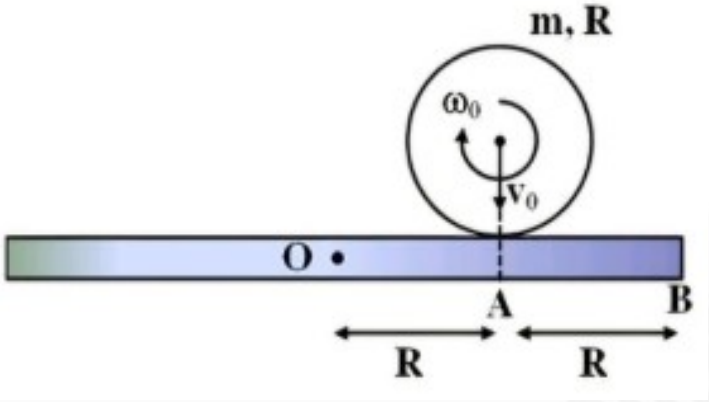
SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A uniform rod of mass M = 3 kg and length $l = 4R = 0.80$ m is hinged at its centre O and located in a vertical plane. A solid ball of mass m = 1 kg and radius R = 0.20 m having initial angular velocity $\omega_0 = 35$ rad/s moving vertically downward with a velocity $v_0 = 10\,m/s$ collides elastically with the rod at a point A which is located at a distance R = 0.20 m from the centre 'O' of the rod as shown in the figure. The coefficient of friction between the ball and the rod is $\mu = 0.5$. Then answer the following questions.

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31. The angular velocity of the rod just after collision is ……..rad/s

32. The angular velocity of the ball just after collision is ……rad/s

PARAGRAPH-II

A particle projected from a point P(0, 0, 600 m) lying in horizontal X – Z plane with intial velocity of $\vec{u}=\left(80\hat{i}+100\hat{j}-60\hat{k}\right)m/s$ at t = 0. It hits a smooth vertical wall in X – Y plane (y-axis in vertically upward direction coefficient of restitution $e=\frac{1}{2}$) $(g=10m/s^2)$

33. Position of the point of impact with the horizontal X-Z plane after collision with the wall is (x, y, z). Find (x + y + z) /100.

34. The ball hits the wall at time t (in sec) when its velocity is making an angle θ (in degrees) with the vertical. Then write θ/t.

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. "$R_H hck^2z^2$" (all letters in usual notation) represents _____ in a single electron species (isoelectronic with H)

A) radius of Bohr's orbit **B)** potential energy of first orbit
C) ionization energy in ground state **D)** velocity of electron in ground state

36. $Fe_xAl_2O_4$ is a metal deficient oxide. If Fe^{+2} and Fe^{3+} are in 1:2 molar ratio, what is x?

A) 0.4 **B)** 0.6 **C)** 0.75 **D)** 0.9

37. How many of following statements are correct

I) CH_3NO_2 have more Pk_a value than $CH \equiv CH$

II) $CH_2(CN)_2$ have more acidic hydrogen than $Ph-\overset{\overset{O}{\|}}{C}-CH_3$

III) Rate of solvolysis of Me_3CBr is more faster in 50% aqueous ethanol than in ethanol in alone.

IV) Naphthalene is relatively more soluble in non polar solvents than Azulene.

V) 2-chloro Ethanoic acid is more acidic than formic acid.

VI) Highly reactive aromatic compounds such as phenol are found to undergo readily nitration even in dilute nitric acid.

A) 5 **B)** 2 **C)** 3 **D)** 4

38. Considering only "particle number " factor, in which of the following "ligand substitution" reactions with one mole reactant (100% yield) maximum 'ΔS' may be observed ?

A) $\left[Co(H_2O)_6\right]^{3+}$ with $NH_3(excess)$

B) $\left[Co(NH_3)_6\right]^{3+}$ with excess 'en'

C) $\left[Co(H_2O)_6\right]^{3+}$ with excess 'glycinate'

D) $\left[Co(ox)_3\right]^{3-}$ with excess 'en'

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Which statement(s) is (are) correct ?

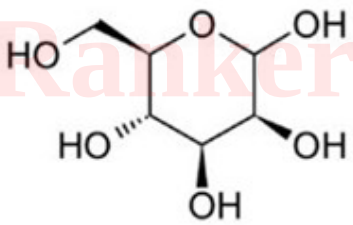
A) During electrolysis of dilute $NaNO_3$, the molar conductance of the medium gradually decreases through loss of water.

B) During electrolysis of aqeous $NaCl$, water is reduced at mercury cathode

C) One litre of 0.1 M $AgNO_3$ on electrolysis with 0.2F charge gives 1:1 theoretical yield of anode and cathodic gaseous products in volume

D) During electrolysis, of $aq.AgNO_3$ with inert electrodes the molar conductance of the medium increases.

40. The correct statement(s) about the pyranose form of a sugar (X) given below is/are:



(X)

A) It exists in two anomeric pyranose forms

B) It reacts with Tollen's reagent to give a silver mirror

C) The penta-O-methyl derivative of (X) is non reducing

D) The penta-O-methyl derivative of (X) is reducing

41. Which of the following comparisions is correct ?

A) $H_3PO_3 > H_3PO_4$ (K_{a_1})

B) $PF_3 > PCl_3$ (ease of hydrolysis)

C) $PH_3 > AsH_3$ (bond angle)

D) $PF_3Cl_2 > PCl_3F_2$ (dipolement)

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SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

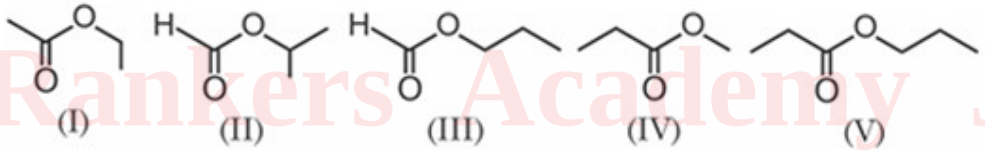
42. How many grams of $RNH_3^+NO_3^-$ (soluble) should be kept in dissolved state in 500 ml aqeous solution such that a salt "AgX" is 1000 times more soluble in this solution than in pure water ? $(K_b \text{ of } RNH_2 = 10^{-12})$

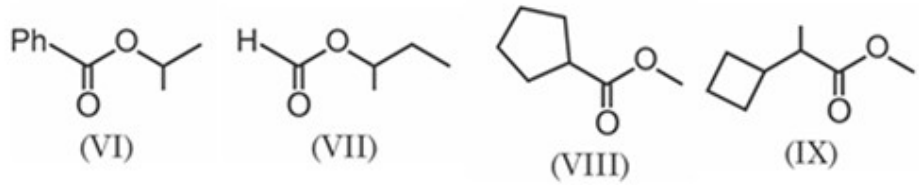
(Molecular weight of $RNH_3NO_3 = 100$) $(K_a \text{ of } HX = 10^{-8})$

43. $\underset{(s)}{3M} + \underset{(g)}{4H_2O} \rightleftharpoons \underset{(s)}{M_3O_4} + \underset{(g)}{4H_2}$

In a closed container at 100^oC finely divided Iron is treated with steam and the equilibrium mixture is formed with 25% relative humidity at 1 atm pressure. Find K_P of the equilibrium.

44. An organic compound reacts with excess of MeMgBr followed by H_2O to give two organic products of which one reacts with NaOI to give yellow precipitate and OTHER DOES NOT. How many of the following compounds can satisfy this condition ?





45. How many of the following are coloured (VIBGYOR) tetrahedral sepecies ?

HgI_4^{2-}, $Cu(CN)_4^{3-}$, $Cu(NH_3)_4^{2+}$, $Zn(NH_3)_4^{2+}$, $PtCl_4^{2-}$,

$Ni(CN)_4^{2-}$, $Pd(CN)_4^{2-}$, MnO_4^{2-}, $Cd(CN)_4^{2-}$, $PbCl_4^{2-}$

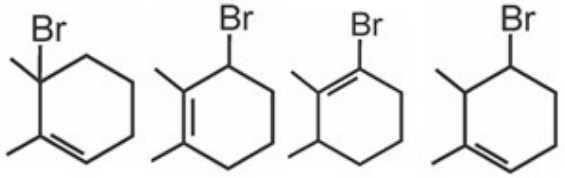
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46. One litre of a 1M solution of nitrate salt of a mono valent metal 'A' is placed in a large vessel made of another divalent metal 'B'. $\left(E^0_{B^{2+}/B} = 0.02V\right)$. After certain period of time 64 grams of 'A' is deposited and stabilised. (At.wt. of 'A' = 80). The reduction potential of $'A^+ / A'$ redox couple in the resultant solution is $a \times 10^{-3}\ volts$. What is $'a'$?

$(\log 2 = 0.3; \frac{RT}{F} = \frac{0.06}{2.303})$

47. What is the sum of position numbers assigned to Bromine atoms while numbering the parent chain in given structures. (As per IUPAC)

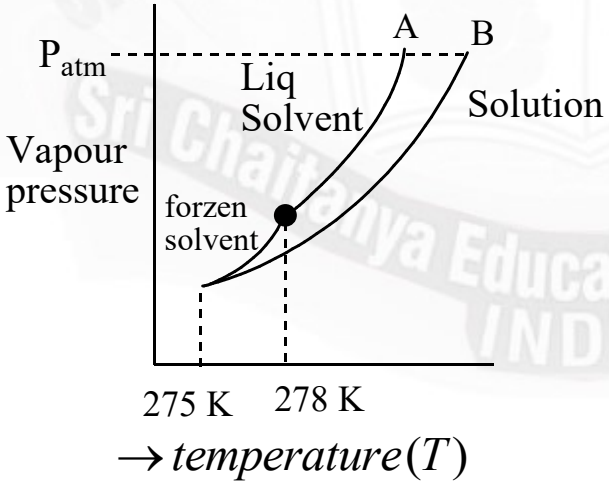


SECTION-IV
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PARAGRAPH-I

Observe the following vapour pressure versus temperature (K) graph given by a solution (B) with organic solvent (A). (Mol.wt.of A = 90)

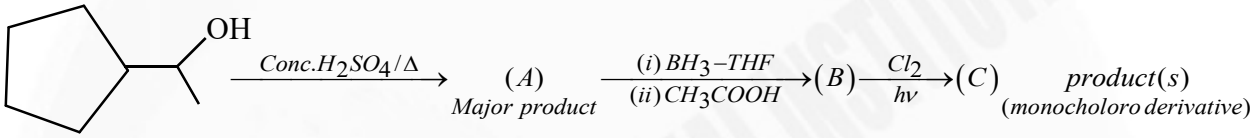


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48. If 2 molal solution (C) (not shown in graph) with 80% dimerised solute starts freezing at $-2.2^{o}C$, then what is the molality of solution 'B' shown in the graph containing same solute with same solvent with 50% dimerisation of solute ?

49. What is the vapour pressure (cm Hg) of solution 'B' at the boiling point of the solvent 'A' ? $(Assume\, X_{solute} = \dfrac{n^{o}_{solute}}{n_{solvent}})$ (M.Wt of A=100)

PARAGRAPH-II



50. Total number of products possible including stereo isomers for given reaction.

$$A \xrightarrow[h\upsilon]{NBS} \overset{X}{(\text{monobrominated products})}$$

51.

$$B \xrightarrow[h\nu]{Br_2} major\ product \xrightarrow[workup]{t-BuO^{-}} \xrightarrow[Me_2S]{O_3} \underset{(cyclic\ product)}{D} + C$$

$$D \xrightarrow[heat]{OH^{-}} E,$$

Degree of unsaturation of 'E'?

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